# Resistors, Capacitors, Inductors

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# Resistors

Resistors follow Ohm's Law:

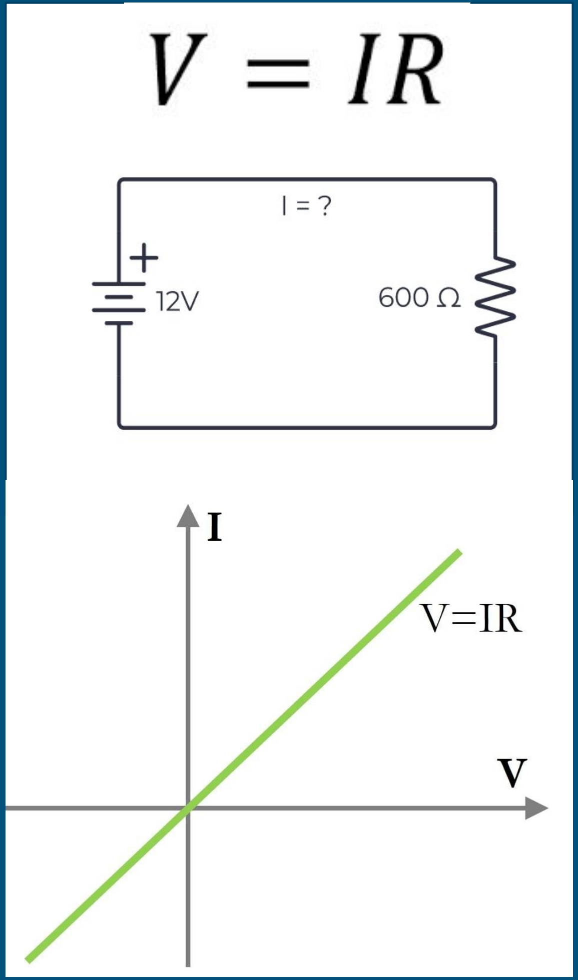
(V)oltage: Electric potential

(I) Current: Flow of electrons

® Resistance: Resistance to flow

Acting as **linear** devices.

**Q**: For a circuit with a 12V voltage and a 600 Ohm resistor, what is the current?



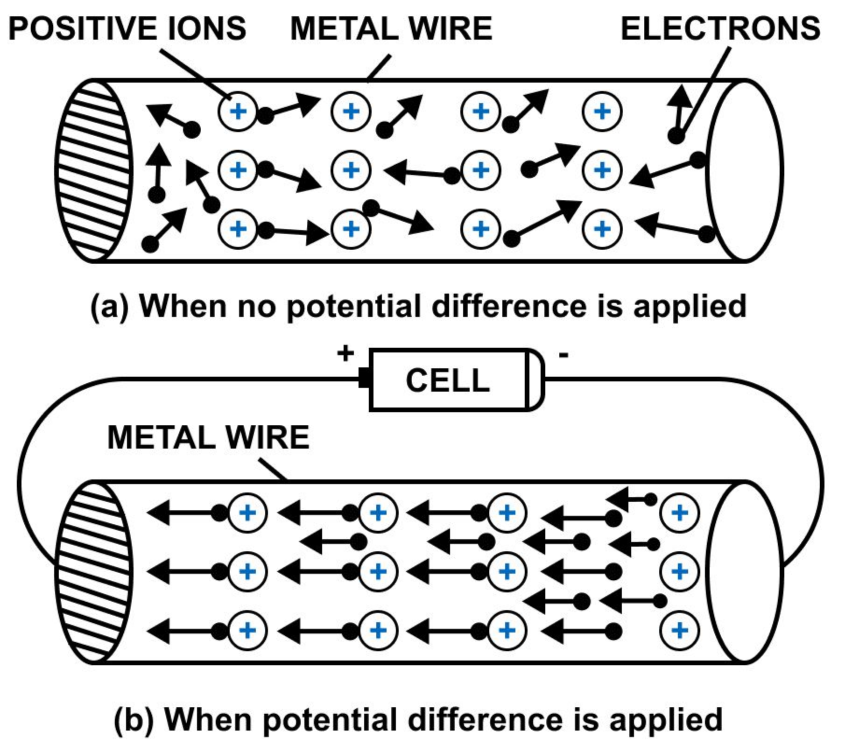
# Resistors

Measure of a material's opposition to flow of electric current.

Units = Ohms = V/A

What might resistance of a material depend on?

$$R = \rho \frac{\ell}{A}$$




(a) When no potential difference is applied

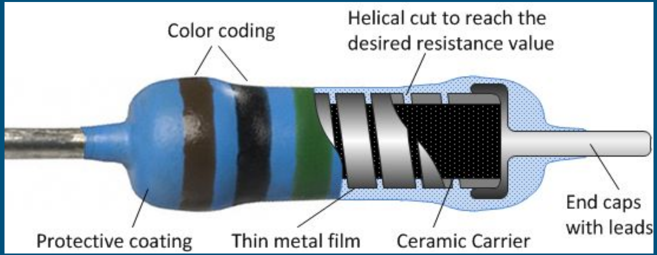
(b) When potential difference is applied

# Resistor - Variable or Fixed

Resistors are used to ‘gate’ the current or voltage through a system. Will use ‘fixed’ value resistors.

But often the ‘responsivity’ or transducer for our detector.

Converting a physical phenomenon to something we can electrically detect.




Fixed: Thin film resistor


Variable: A thermistor

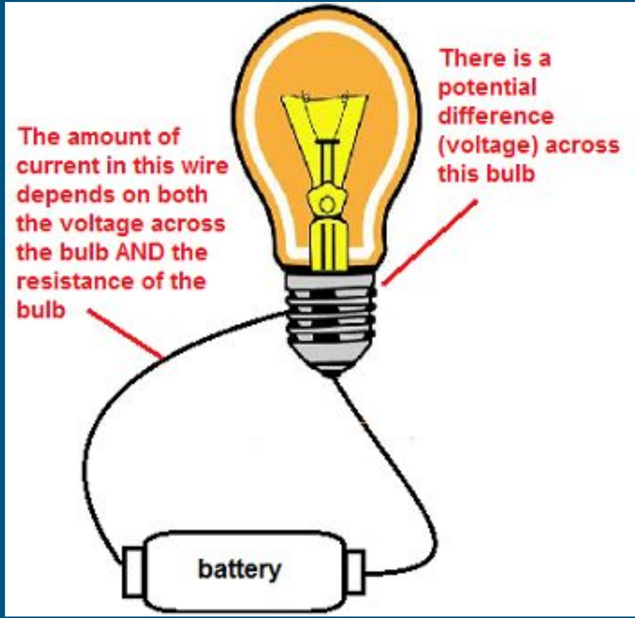
# Power

When voltage and current run through a component, that is electrical energy being transferred - **power**.

For a resistive component, some power will always be converted to heat.

Different devices will convert power differently.

$$P = IV$$

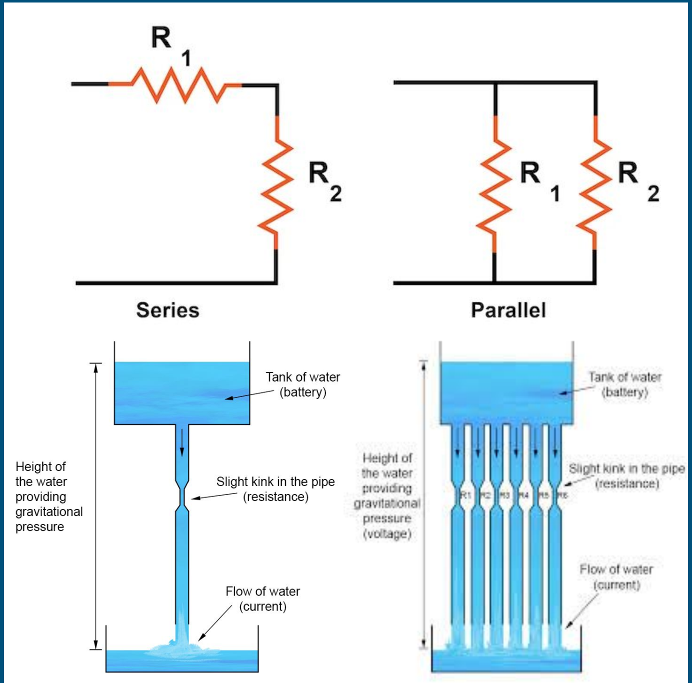


# Parallel and Series Circuits

How we hook up components affects the flow of current. Because of Ohm’s law, that will change the voltage across each component.

Series: Only 1 path for current, so current is the same through both devices. Voltage wil

Parallel: Current can flow down both paths. Voltage the same for



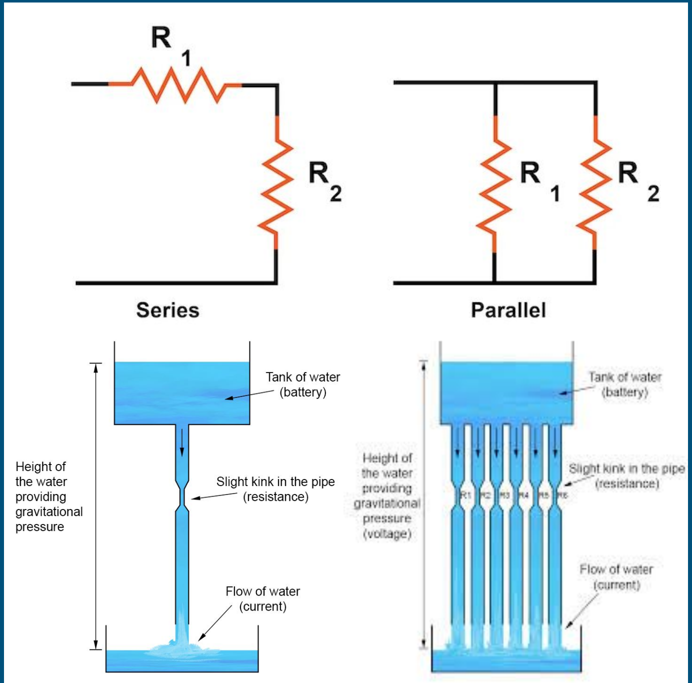
# Parallel and Series Circuits

Series: Only 1 path for current.

- Current is equal for R1 and R2.
- Voltage is different for R1 and R2.

Parallel: Current can flow down both paths.

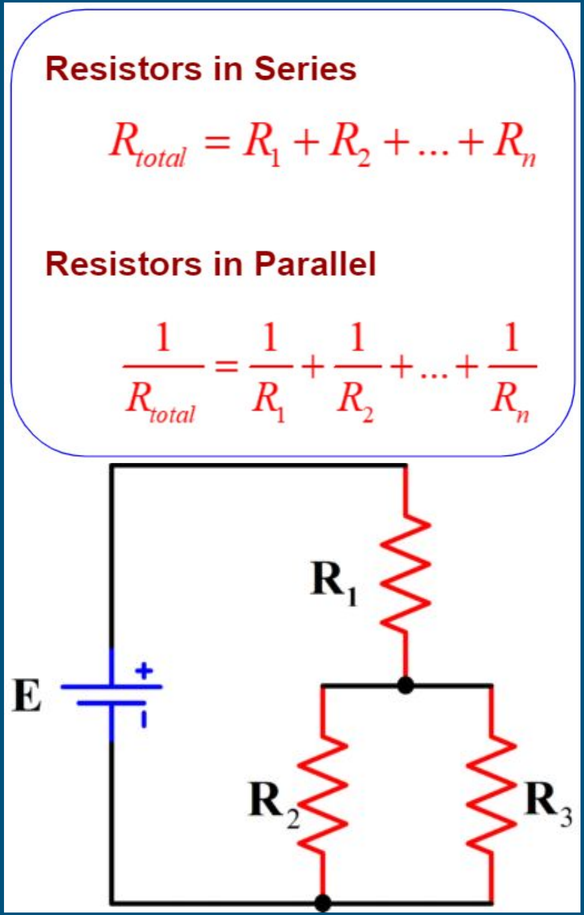
- Current is different for R1 and R2
- Voltage is the same for R1 and R2.



# Thevenin Equivalence

We can reduce circuits down to simpler versions to focus on a specific device.

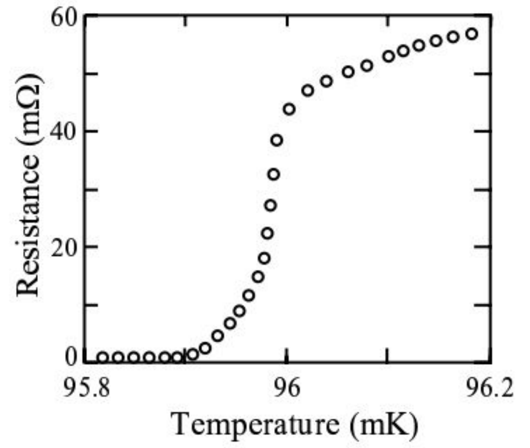
We can use this to understand what will happen to a new device if we connect it to a circuit.

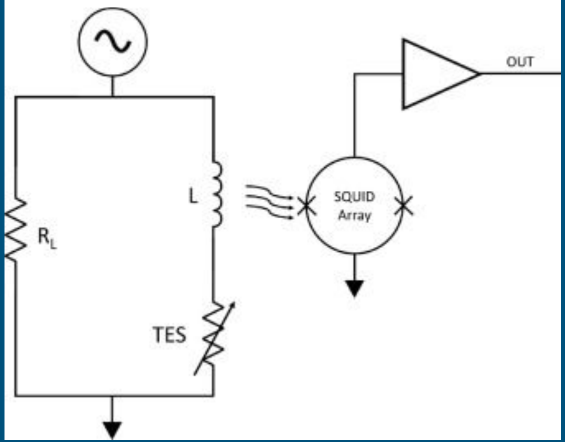


# Voltage Divider - TES Biasing

To keep a TES within the transition, we need to keep it near the critical temperature. Apply a ‘bias’ voltage, providing constant voltage to heat up the TES.

But if a TES resistance changes depending on temperature, how do we apply a constant voltage?



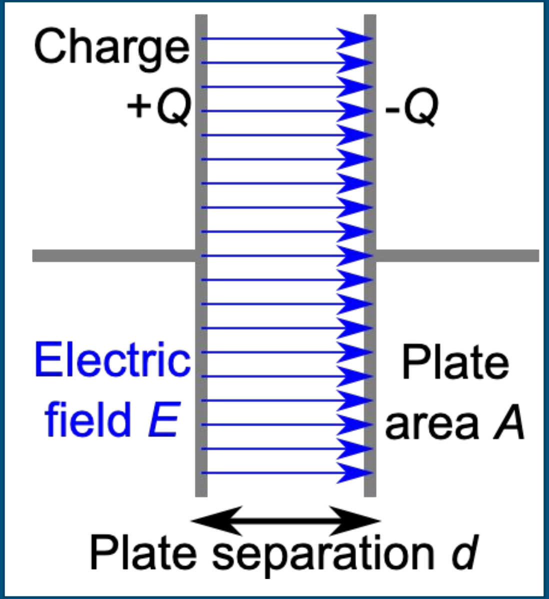



# Capacitors

A capacitor is a passive device consisting of two plates at a uniform separation.

$$E = \frac{q}{\varepsilon A} \qquad v = Ed = \frac{qd}{\varepsilon A}$$

$$i = \frac{dq}{dt} = \frac{d}{dt}\left(\frac{\varepsilon A}{d} v\right) = \frac{\varepsilon A}{d}\frac{dv}{dt} = C\frac{dv}{dt}$$

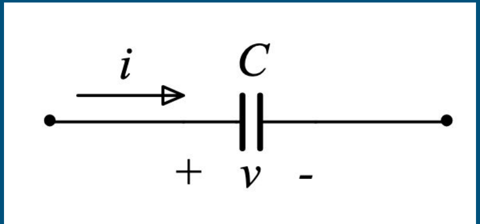


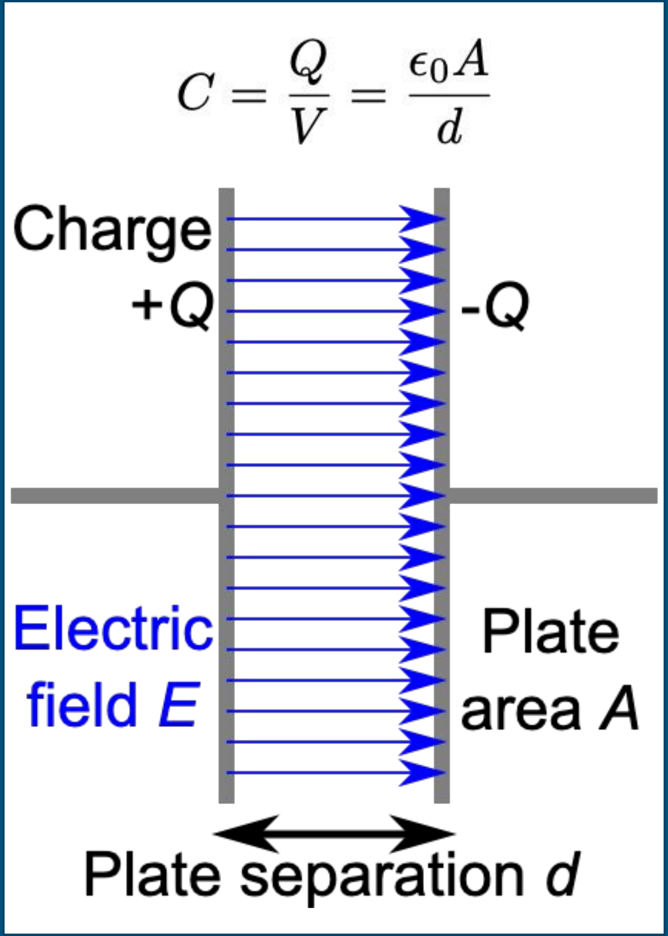
# Capacitance

**Capacitance** determines how much charge a capacitor can hold at a given voltage.

In the simplest case, the capacitance is determined **only** by the geometry of the plates.

How can we increase the capacitance?



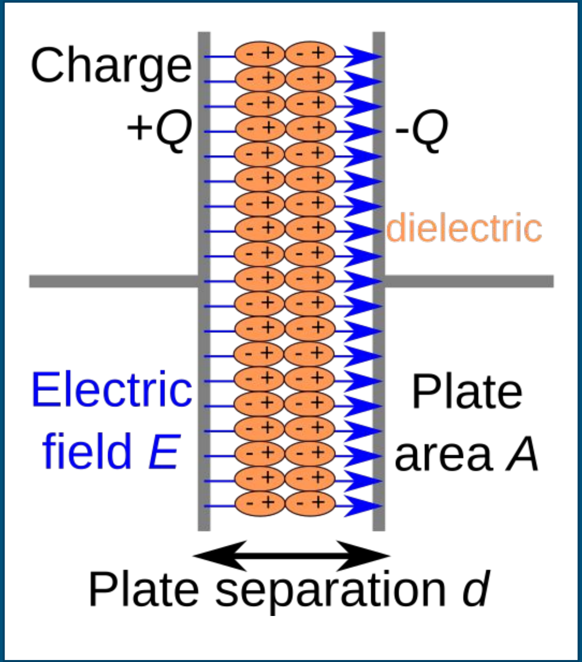
# Capacitors and Dielectrics

**A dielectric** is a material that acts like an electric insulator. By inducing polarization on the medium, this reduces the overall field within the dielectric itself.

Electric field reduced -> voltage for same charge reduced.

$$v = Ed \qquad C = \frac{q}{v} \qquad C = \kappa \frac{\epsilon_0 A}{d}$$

Capacitor can store same amount of charge at a lower voltage!



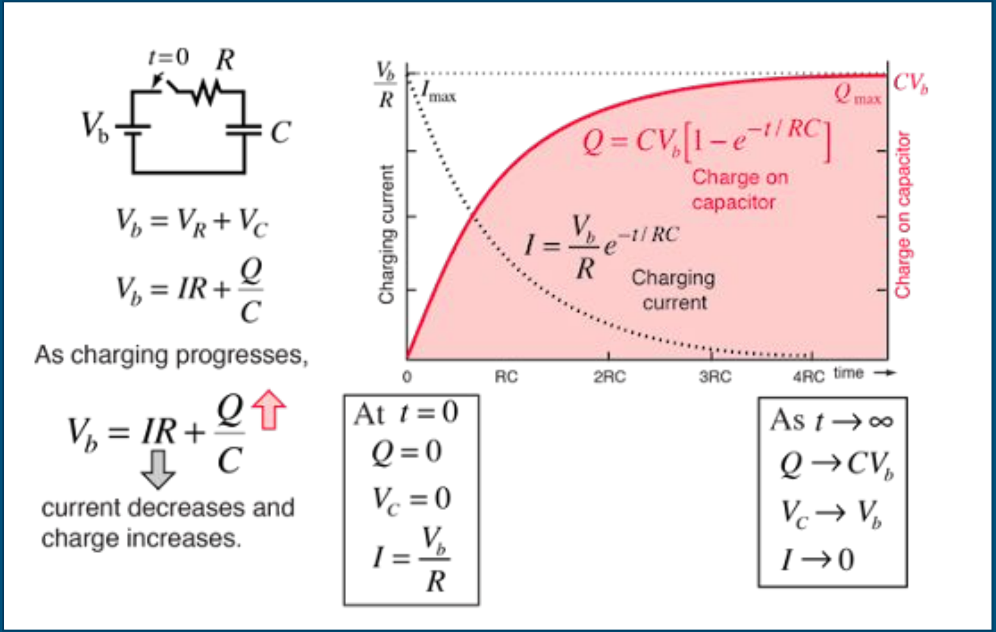
# Capacitors in Time Domain

By solving differential equations:

$$i = C\frac{dv}{dt} = \frac{dq}{dt} \quad v = R\frac{dq}{dt} + \frac{q}{c}$$

$$q = Ae^{-pt} + B$$

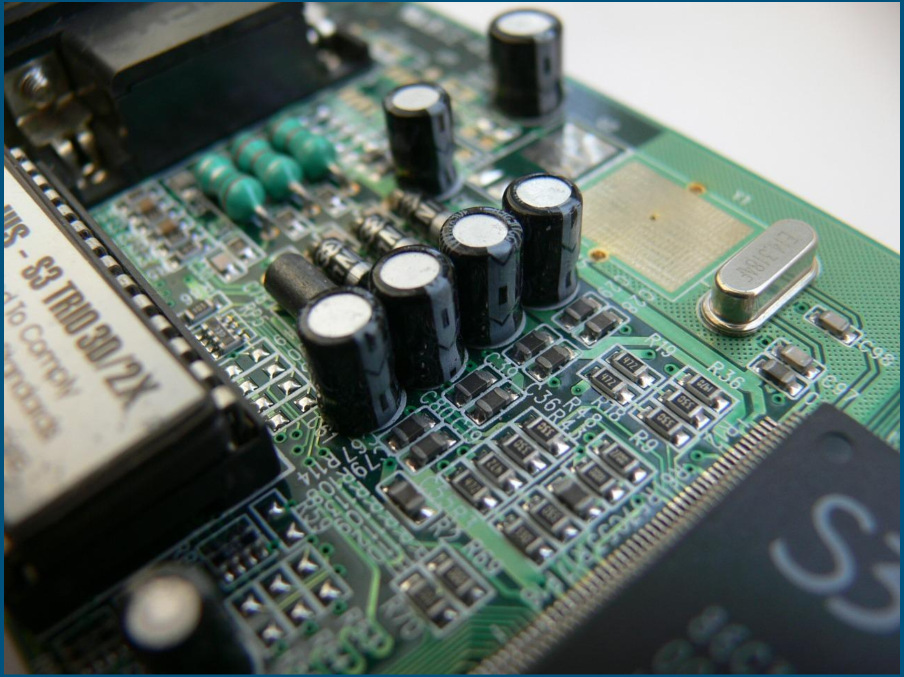
$$q = CV_b[1 - e^{-t/RC}]$$



# Capacitors - Energy Storage

Used on many circuit boards to store temporary power. Charged up while operating.

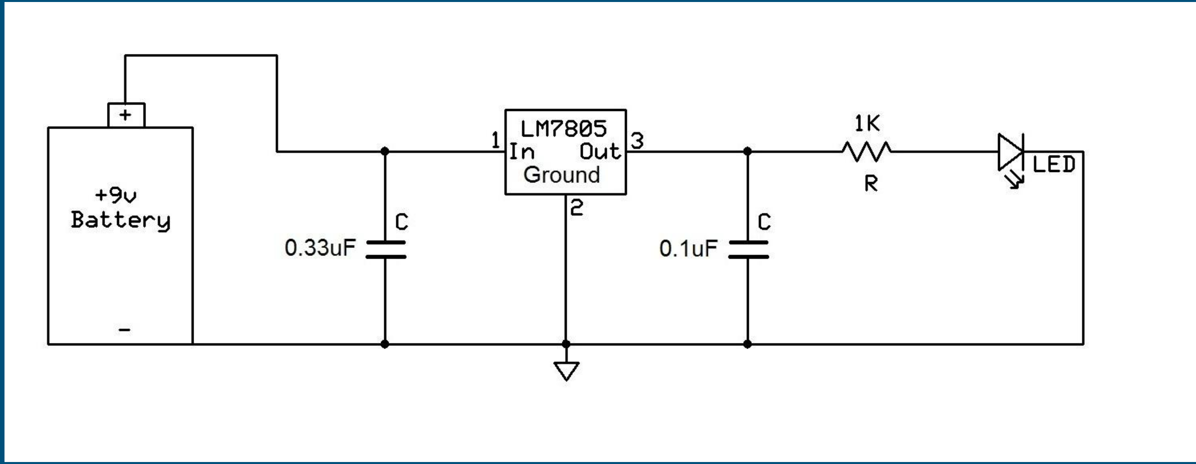
If a power outage occurs, capacitors provide enough power for the computer to shut down safely.



# Capacitors - Power Conditioning

A voltage regulator ‘cleans’ an inputted voltage. Power from battery can be inconsistent.

Capacitors here will charge up, and if battery power ‘sags’, will discharge to maintain original voltage.



# Types of Capacitors

Ceramic: Unpolarized, cheap, compact. Good at high frequencies. For coupling/filters. Go to choice!

Plastic film: rolled plastic film between metal layers. I rarely use over ceramic.

Electrolytic: big, polarized, using an electrolyte plate to get more capacitance. Try not to use, can catch fire, will short if used wrong! Good for DC power storage.

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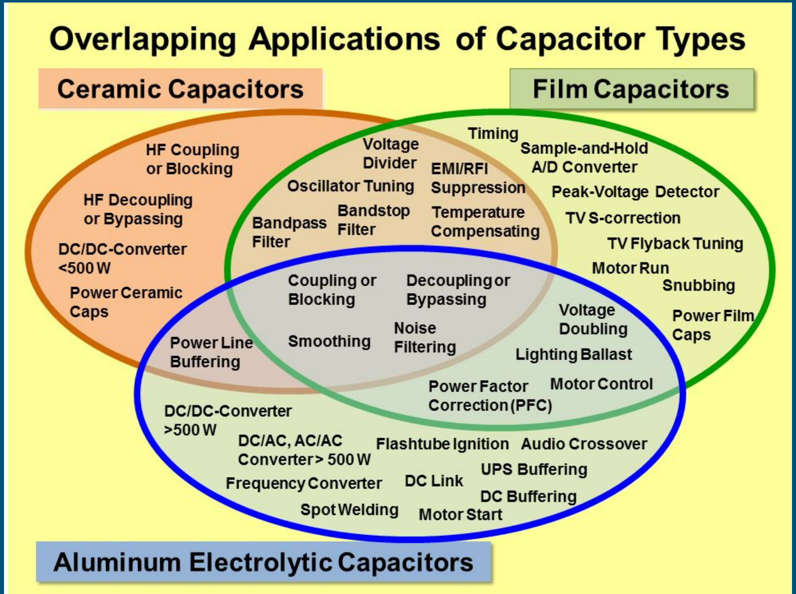
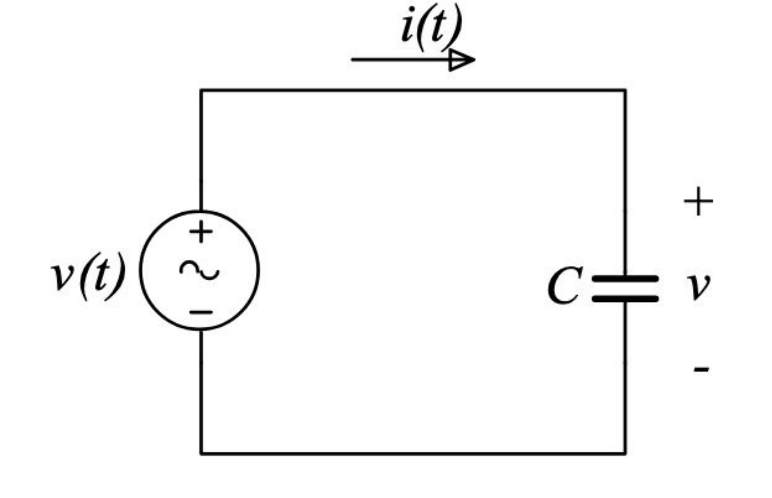


# AC and DC Behaviour of Capacitors

DC Case: a capacitor acts like a short. Once charged it is an open circuit.

What does a capacitor do in AC?



# AC Behaviour of Capacitors

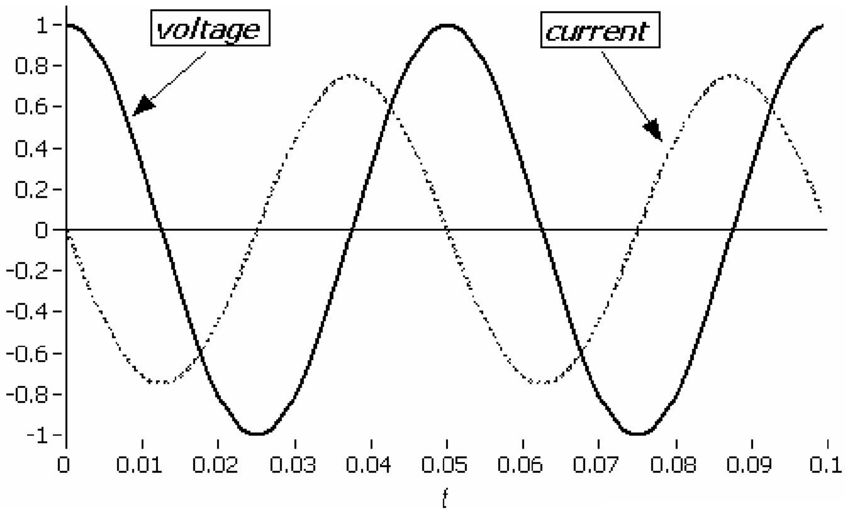
If the voltage is:

$$v(t) = A\cos(\omega t)$$

$$i(t) = C\frac{dv}{dt}$$

$$= -C\,A\,\omega\sin(\omega t)$$

$$= C\,\omega A\cos\left(\omega t + \frac{\pi}{2}\right)$$

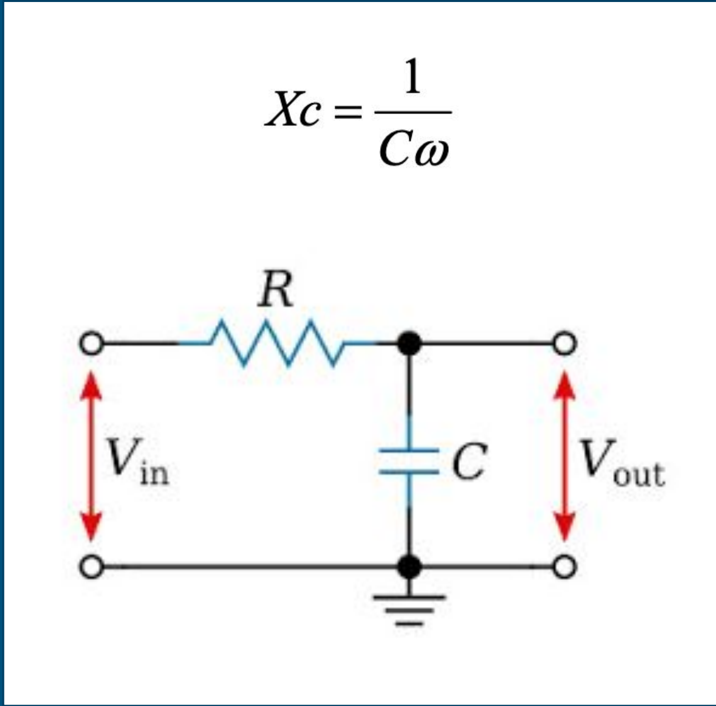



If we take the ratio of the peak voltage to peak current:

$$Xc = \frac{1}{C\omega}$$

# Frequency dependent Behaviour

Xc is the what we call the ‘impedance’ of a capacitor, how much it resists flow. We are now starting to see components that act differently on different frequencies. It has high ‘impedance’ at high frequencies.



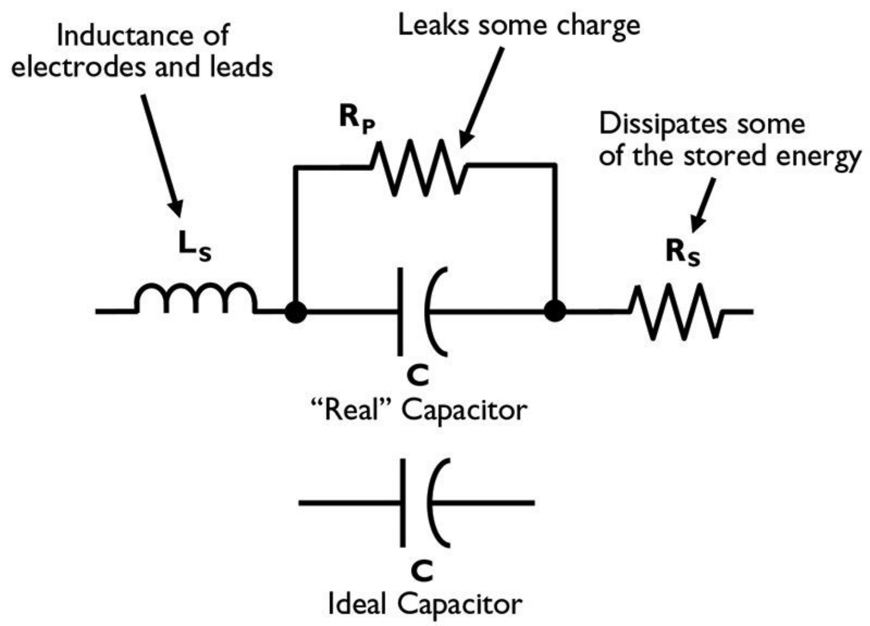

What does this circuit do then? Will it let more high frequency through or low?

We are approaching talking about transfer functions.

# Non-Idealities

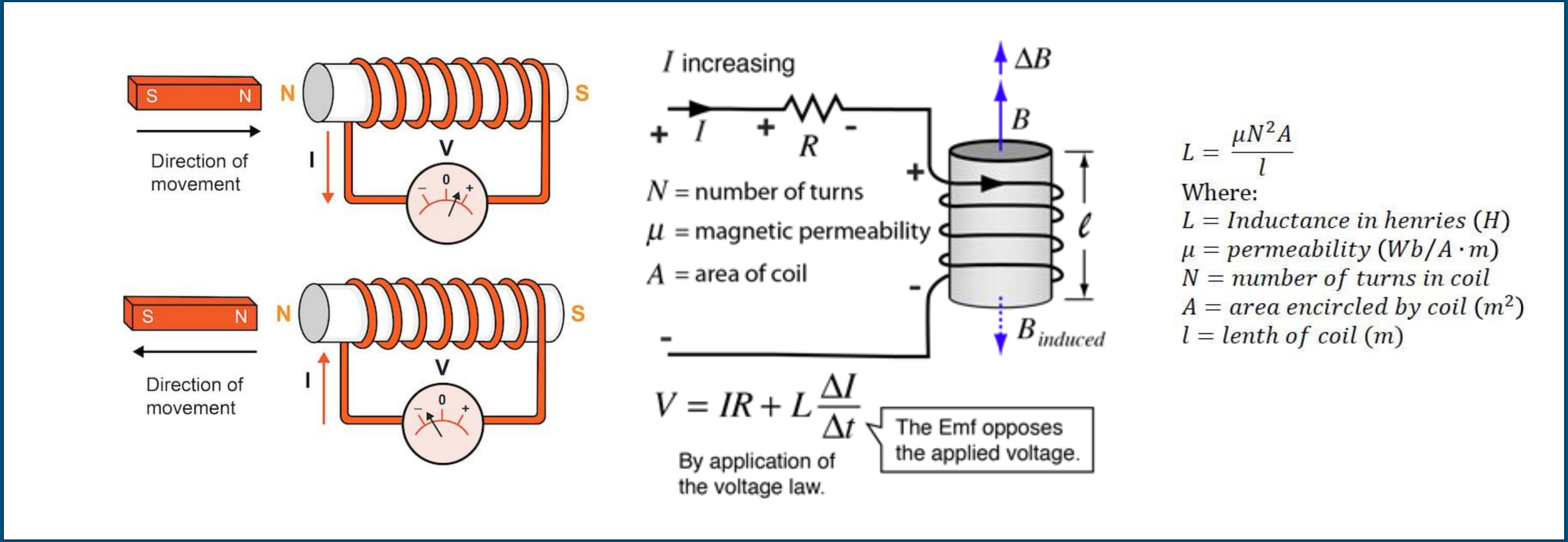
Reality is disappointing…

A capacitor in reality will have inherent losses and parasitic inductance and resistance. These will matter in precision measurements.



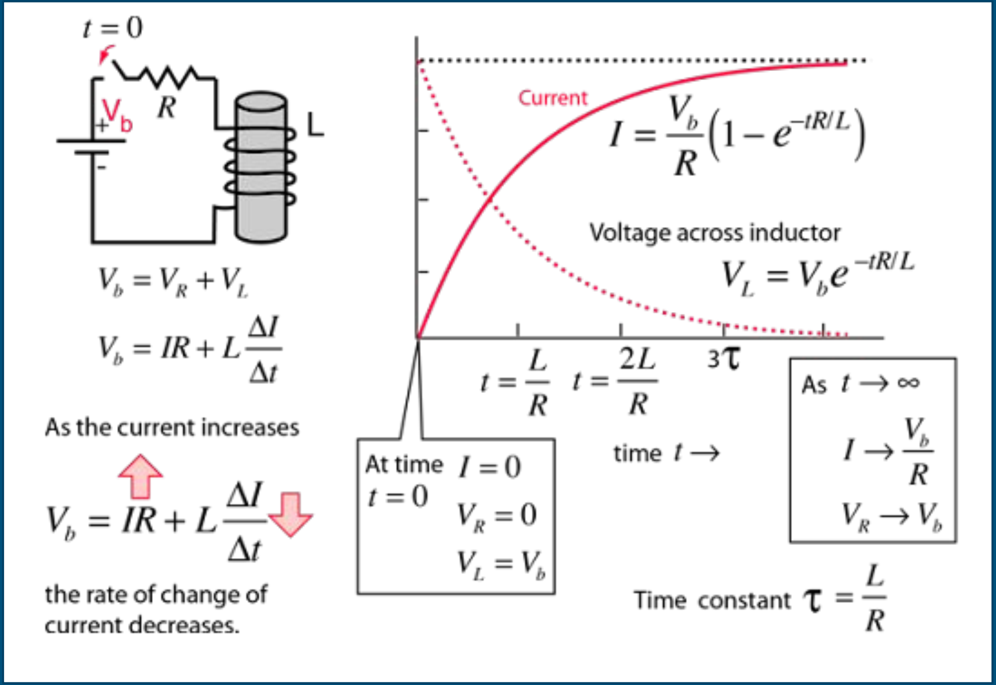
# Inductors

Inductors hold **magnetic energy**.



# Inductor In Time Domain

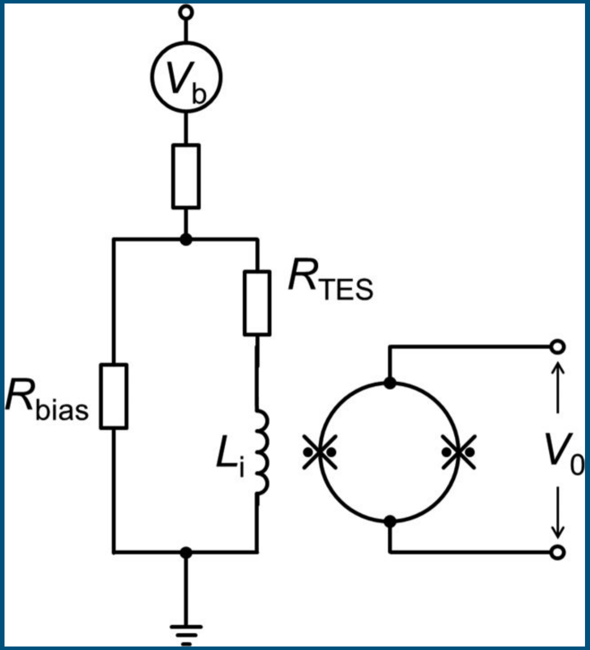
We can solve a similar system of equations to solve the time domain/transient behaviour of inductors.



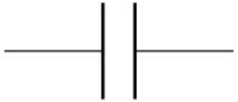
# Applications of Inductors

- Somewhat less common, mostly used in AC circuits for filtering purposes

- Can be used for coupling to SQUID readouts for TES readout!



# Capacitors and Inductors Summarized

| | **Capacitor** | **Inductor** |
|---|---|---|
| Symbol |  |  |
| Stores energy in | electric field | magnetic field |
| Value of component (unit) | capacitance, $C$ (farad, F) | inductance, $L$ (henry, H) |
| I–V relationship | $i = C\dfrac{dv}{dt}$ | $v = L\dfrac{di}{dt}$ |
| At steady state, looks like | open circuit | short circuit |

# Thank you!

Questions?